The documentation and process conversion measures necessary to comply with this revision shall be completed by 22 January 2019.

INCH-POUND

MIL-PRF-19500/429P
w/AMENDMENT 1
22 October 2018
SUPERSEDING
MIL-PRF-19500/429P
9 March 2018

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY, TYPES 1N5615, 1N5617, 1N5619, 1N5621, 1N5623, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, fast recovery power rectifier diodes that are hermetic glass encapsulated. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device as specified in MIL-PRF-19500. Two levels of product assurance (JANHC and JANKC) are provided each unencapsulated device type.

1.2 Package outlines and die topography. The device package for the encapsulated device type are as follows: Axial in accordance with figure 1, surface mount version US (D-5A) in accordance with figure 2, and surface mount version UMC in accordance with figure 3. The dimensions and topography for JANHC and JANKC unencapsulated die are as follows: A version die in accordance with figure 4, B version die in accordance with figure 5, and C version die (1N5615) in accordance with figure 6.

1.3 Maximum ratings. Unless otherwise specified T_C = +25°C.

Col. 1	Col. 2	Col. 3		Col. 4	Col. 5	Col. 6	Col. 7	Col. 8	Col. 9	Col. 10	Col. 11	Col. 12
Types (1)	V_{RWM}	I_O (2) (3)		t_{rr}	T_{STG} T_J	I_{FSM}	Barometric pressure (reduced) (4)	$R_{\theta JL}$	$R_{\theta JEC}$	$R_{\theta JX}$	$R_{\theta JA}$ (7)	$R_{\theta JSP}$ (7)
		T_A = +55°C	T_A = +100°C			T_A = +100°C I_O = 750 mA dc V_{RWM} = rated t_P = 8.3 ms		L = .375 inch (9.53 mm) (5)	(6)			
	V(pk)	A dc	mA dc	ns	°C	A(pk)	mmHg	°C/W	°C/W	°C/W	°C/W	°C/W
1N5615, UMC	200	1	750	150	-65	25	N/A	38	13	115	44	9.15
1N5617	400	1	750	150		25	8	38	13	115		
1N5619	600	1	750	250	to	25	8	38	13	115		
1N5621	800	1	750	300		25	33	38	13	115		
1N5623	1,000	1	750	500	+175	25	33	38	13	115		

See notes on next page.

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1.3 Maximum ratings - Continued.

(1) Electrical characteristics for US suffix devices are identical to the corresponding non-suffix device.
(2) Derate linearly from 1.0 A at T_A = +55°C to 0.75 A at +100°C. Derate linearly from 0.75 A to 0 A between +100°C and +175°C. See figure 7 for derating curves.
(3) For the 1 amp rating at 55°C ambient or 0.75 amp rating at 100°C ambient, these I_O ratings are for thermal mounting methods (PC boards or other) where thermal resistance from mounting point to ambient is still sufficiently controlled where $T_{J(MAX)}$ in 1.3 is not exceeded. This equates to $R_{\theta JX} \leq 115^{\circ}C/W$ as shown. Also see application notes in 6.6.1.
(4) Barometric pressure: 1N5617 and 1N5619 is 8 mm Hg (100,000 feet); 1N5621 and 1N5623 is 33 mm Hg (70,000 feet).
(5) See figure 8.
(6) See figure 9.
(7) 1N5615UMC only, PCB Size: 1.5" x 1.5", FR4, 1.0 oz Copper. See figures 10 and 11.

1.4 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.4.1 JAN certification mark and quality level.

1.4.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.4.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.4.2 Device type. The designation system for the device types of semiconductors covered by this specification sheet are as follows.

1.4.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

1.4.2.2 Second number symbols. The second number symbols for the semiconductors covered by this specification sheet are as follows: "5615", "5617", "5619", 5621", and "5623".

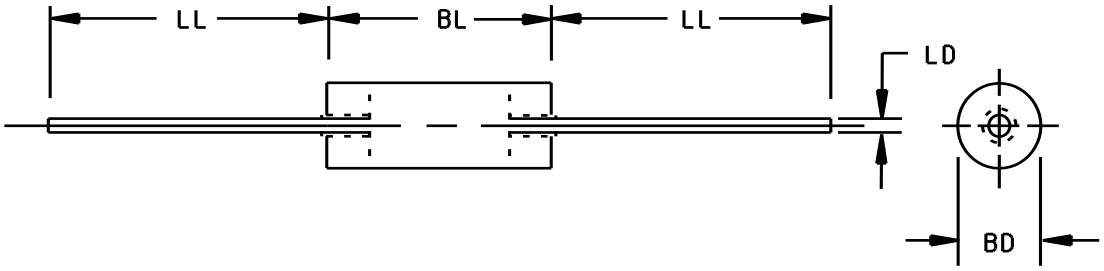
1.4.3 Suffix symbols. The following suffix symbols are incorporated in the PIN as applicable.

	A blank suffix symbol indicates a through-hole mount axial package (see figure 1).
US	Indicates a surface mount package with square endcaps (see figure 2).
UMC	Indicates a surface mount package (1N5615 only, see figure 3).

1.4.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.4.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A", "B", and "C" (see figures 4, 5, and 6 and 6.5.2).

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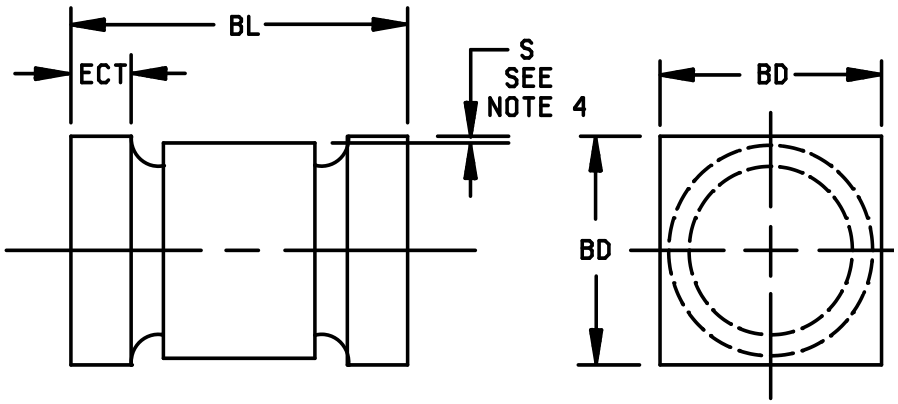


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min Min	Max	Min	Max	
BD	.065	.110	1.65	2.79	3
LD	.026	.033	.66	.84	
BL	.130	.225	3.30	5.71	4
LL	1.00	1.30	25.40	33.02	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimension BD shall be measured at the largest diameter.
4. Dimension BL shall include the entire body including slugs and sections of the lead over which the diameter is uncontrolled. This uncontrolled area is defined as the zone between the edge of the diode body and extending onto the leads .050 inch (1.27 mm)
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (for non-US suffix devices only) (axial lead).

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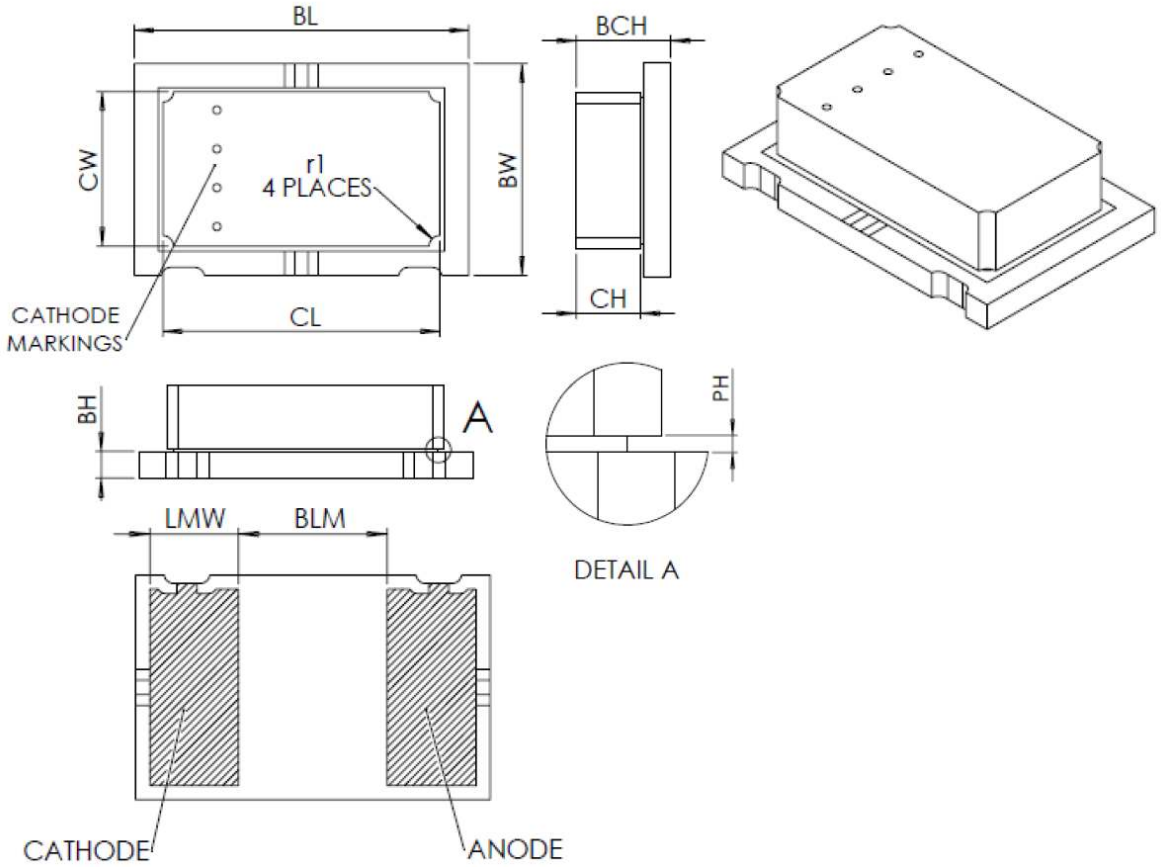


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.168	.200	4.27	5.08
ECT	.019	.028	0.48	0.71
S	.003		0.08	
BD	.091	.103	2.31	2.62

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimensions are pre-solder dip.
4. Minimum clearance of glass body to mounting surface on all orientations.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions for types 1N5615US, 1N5617US, 1N5619US, 1N5621US, and 1N5623US, (surface mount devices) (D-5A).

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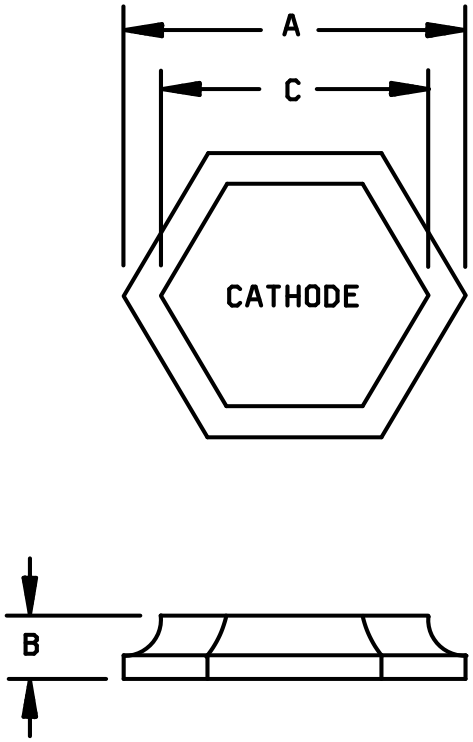


Ltr.	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
CL	.199	.211	5.06	5.36
CW	.109	.120	2.76	3.06
BL	.246	.250	6.25	6.35
BW	.155	.159	3.94	4.04
r1	.002	.014	0.05	0.35
BCH	.052	.070	1.32	1.78
CH	.033	.045	0.84	1.14
BH	.018	.022	0.46	0.56
BLM	.102	.106	2.59	2.69
LMW	.060	.064	1.52	1.63
PH	.001	.003	0.03	0.07

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas denote metallized areas.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions of 1N5615UMC surface mount.

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A - version

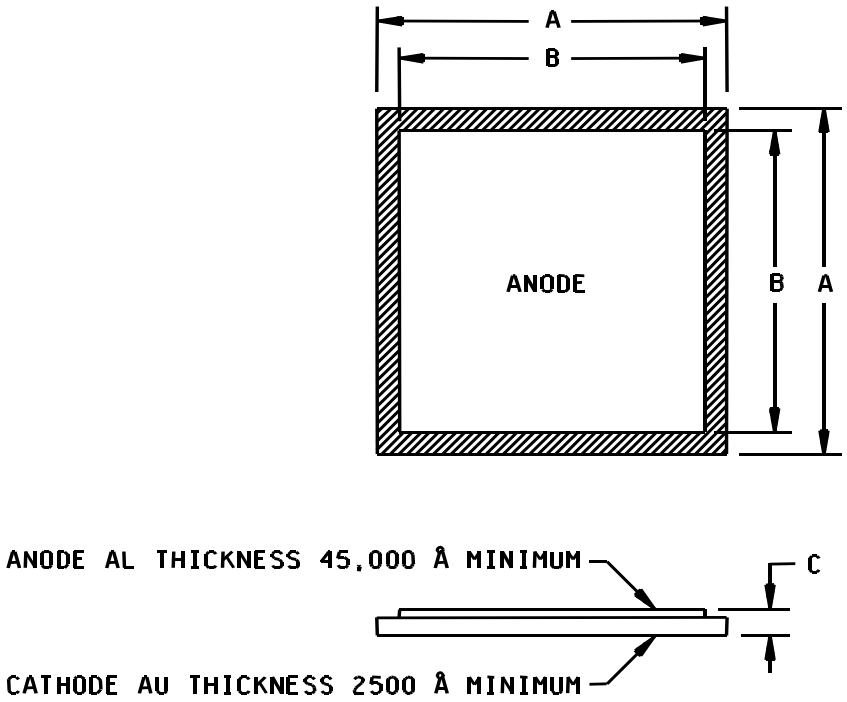
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.047	.053	1.19	1.35
B	.007	.011	0.18	0.28
C	.033	.037	0.84	0.94

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Gold 10,000 Å minimum.
   Back metal: Gold 4,000 Å minimum.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 4. Physical dimensions, JANHCA and JANKCA die.

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B - version



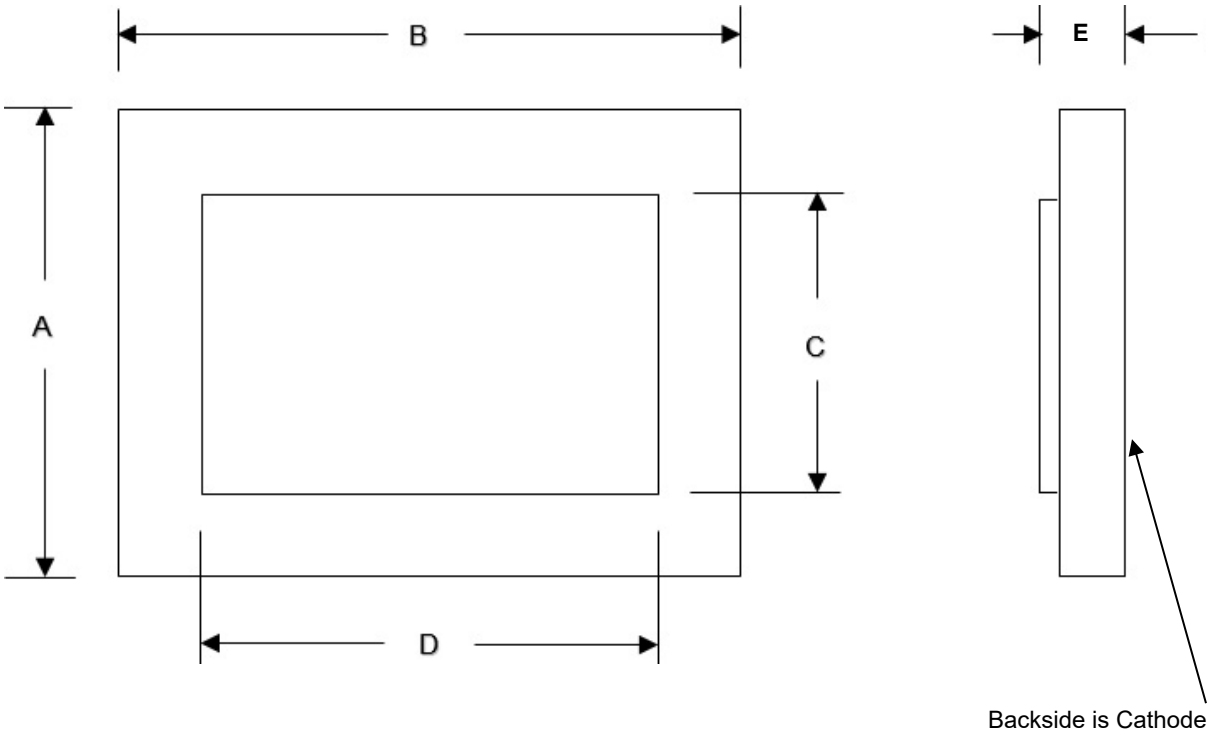
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	1.00	1.09
B	.021	.025	0.53	0.64
C	.008	.012	0.20	0.31

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Aluminum (anode) 45,000 Å minimum.
   Back metal: Gold (cathode) 2,500 Å minimum.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 5. Physical dimensions, JANHCB and JANKCB die.

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C-version



Ltr	Dimensions 1N5615			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.040	.050	1.016	1.270
B	.055	.065	1.397	1.651
C	.024	.034	0.610	0.864
D	.039	.049	0.991	1.245
E	.003	.013	0.076	0.330

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Anode is aluminum at 30,000 Å ± 5,000 Å.
4. Cathode is gold at 6,500 Å ± 1,500 Å.

FIGURE 6. Physical dimension, JANHCC and JANKCC die.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

EC	End-cap.
US	Short-body unleaded or surface mounted diodes (square end-caps).
UMC	surface mounted diode with ceramic lid

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (axial lead), figure 2 surface mount (D-5A), figure 3 surface mount (UMC), figures 4, 5, and 6 (JANHC and JANKC).

3.4.1 Lead finish. Unless otherwise specified, lead or end-cap finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. When solder alloy is used for finish the maximum lead temperature is limited to 175°C maximum. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2).

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3.5 Diode construction. These devices shall be constructed utilizing non-cavity double plug construction with high temperature metallurgical bonding between both sides of the silicon die and terminal pins. Metallurgical bonds shall be in accordance with the requirements of category I, appendix A, MIL-PRF-19500. No point contacts are allowed. Silver button dumet design is prohibited. US version devices shall be structurally identical to the non-surface mount devices except for lead terminations. The die for the 'UMC' devices shall be eutectically mounted and wire bonded in a ceramic package.

3.6 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6.1 Marking of US version. For US version only, all marking except polarity indicator (see 3.6.2) may be omitted from the body, but shall be retained on the initial container.

3.6.2 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for US and UMC suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end or a contrasting color band may be used.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or requalification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot to this revision to maintain qualification.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with appendix G of MIL-PRF-19500 and as specified herein.

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4.3 Screening (JANS, JANTX, and JANTXV levels). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table E-IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTXV and JANTX level
1a 1b	Required Required	Not required Required (JANTXV only)
2	Not required	Not required
3a (1) 3c	Required Thermal impedance (see 4.3.1)	Required Thermal impedance (see 4.3.1)
4	Not applicable	Not applicable
5	Applicable to UMC devices only	Not applicable
6	Not applicable	Not applicable
8	Required	Not required
9	Required I_{R1} and V_F	Not required
10	Method 1038 of MIL-STD-750, condition A	Method 1038 of MIL-STD-750, condition A
(2) 11	I_{R1} and V_F, $\Delta I_{R1} \leq$ 100 percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc	I_{R1} and V_F
12	Required, see 4.3.2	Required, see 4.3.2
(2) (3) 13	Subgroups 2 and 3 of table I herein: $\Delta I_{R1} \leq$ 100 percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc. Scope display evaluation (see 4.5.4)	Subgroup 2 of table I herein: $\Delta I_{R1} \leq$ 100 percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc. Scope display evaluation (see 4.5.4)
15	Required	Not required
16	Required	Not required

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.
(2) For JANTX and JANTXV devices, ΔV_{F2} may be omitted if thermal impedance is performed, unless irradiation is used to reduce the carrier lifetime.
(3) $Z_{\theta JX}$ is not required in screen 13, if already previously performed.

4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD 750 as applicable, using the guidelines in that method for determining I_M, I_H, t_H, and K factor where appropriate). Measurement delay time (t_{MD}) = 70 µs max. The thermal impedance limit used in screen 3c and table I, subgroup 2 shall be set statistically by the supplier.

4.3.2 Free air power burn-in conditions. Power burn-in conditions are as follows (see 4.5.3 and 4.5.3.1): $I_{O(min)}$ = rated I_O (see 1.3 herein); Test conditions in accordance with method 1038 of MIL-STD-750, condition B. Use method 3100 of MIL-STD-750 to measure T_J. Adjust I_O or T_A to achieve the required T_J. T_J = 135°C minimum. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.3 Screening (JANHC and JANKC). Screening of die shall be in accordance with appendix G of MIL-PRF-19500. As a minimum, die shall be 100-percent probed to ensure compliance with table I, subgroup 2. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. The $Z_{\theta JX}$ end-point shall be derived by the supplier and approved by the qualifying activity. This $Z_{\theta JX}$ end-point shall be documented in the qualification report.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIA (JANS) and table E-VIB (JANTX and JANTXV) of MIL-PRF-19500, appendix E, and as follows. Delta measurements shall be as specified in table III herein.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500. For B5, if a failure occurs, resubmission shall be at the test conditions of the original sample.

Subgroup	Method	Condition
B3	1056	0°C to +100°C, 25 cycles. Does not apply to UMC diodes
B3	1051	-55°C to +175°C, 100 cycles.
B3	4066	Condition A, I_{FSM} = rated I_{FSM} (see col. 6 of 1.3); ten surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 750 mA, V_{RWM} = rated V_{RWM} (see col. 3 of 1.3) T_A = 100°C.
B4	1037	I_O = 1 A minimum (see col. 3 of 1.3); V_R = rated V_{RWM} (see col. 2 of 1.3, 4.5.3 and 4.5.3.1); 2,000 cycles.
B5	1027	I_O = 1 A minimum (see column 3 of 1.3), apply V_R = rated V_{RWM} (see column 2 of 1.3), adjust I_O or T_A to achieve T_J = +175°C minimum. t = 1,000 hours, n = 45, c = 0. Temporary leads may be added for surface mount devices. For irradiated devices, include t_{rr} as an end-point measurement.
B8	4065	Peak reverse power, see table II, subgroup 8 herein. $P_{RM} \geq 318$ W for square wave in accordance with method 4065 of MIL-STD-750, ($P_{RM} \geq 500$ W for half-sine wave). Test shall be performed on each sublot; sampling plan n = 10, c = 0, end-points, see 4.4.2.

4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	I_O = 1 A minimum adjust I_O or T_A to achieve the required T_J of +150°C minimum. Apply V_R = rated V_{RWM} (see col. 2 of 1.3), f = 50-60 Hz (see 4.5.3 and 4.5.3.1). T_A = 55°C (max). For irradiated devices, include t_{rr} as an end-point measurement.
B5		Not applicable.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500. Delta limits shall be in accordance with table III herein.

4.4.3.1 Group C inspection, appendix E, table E-VII of MIL-PRF-19500.

	Subgroup	Method	Condition
	C2	1056	0°C to +100°C, 10 cycles, Does not apply to UMC diodes.
	C2	1051	-55°C to +175°C, 25 cycles.
*	C2	2036	Tension: Test condition A; weight = 12 pounds (5.44 Kg); t = 15 seconds. Lead fatigue: Test condition E; weight 1 pound (0.45 Kg). NOTE: Lead fatigue is not applicable for US devices. Does not apply to UMC diodes.
*	C2	2038	US devices – Tension: Test condition B; weight = 12 pounds (5.44 Kg); t = 15 seconds. Does not apply to UMC diodes.
	C5	4081	$R_{\theta JL}$ (maximum) ≤ 38°C/W; L = .375 inch (9.53 mm), (see 4.3.1). For surface mount devices (US version), $R_{\theta JEC}$ ≤ 13°C/W, For surface mount devices (UMC version), $R_{\theta JSP}$ ≤ 9.15°C/W.
	C6	1026	I_O = 1 A, minimum; adjust I_O or T_A to achieve T_J = +150°C minimum. Apply V_R = rated V_{RWM} (see col. 2 of 1.3), f = 50-60 Hz (see 4.5.3 and 4.5.3.1). For irradiated devices, include t_{rr} as an end-point measurement.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein; except, $Z_{\theta JX}$ need not be performed. See table III for delta limits when applicable.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature T_A of +25°C ±3°C.

4.5.3 Burn-in and life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees.

4.5.3.1 Free air burn-in and life tests. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the full P_t (minimum) and that the minimum applied voltage, where applicable, is maintained through out the burn-in period. Suppliers must report the (heat sinking), thermal resistance of their mounting conditions in all life test and burn-in boards to the qualifying activity.

4.5.4 Scope display evaluation. Scope display evaluation shall be stable in accordance with method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display in table I, subgroup 4 shall be performed on a scope. Reverse current (I_{BR}) over the knee shall be 500 μA peak.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3101	See 4.3.1	$Z_{\theta JX}$			°C/W
Forward voltage	4011	Condition B, I_F = 3 A dc (pulsed), (see 4.5.1) t_p = 300 µs; 2 percent maximum duty cycle	V_F	0.8	1.6	V_{pk}
Reverse current leakage	4016	DC method	I_{R1}			
1N5615, US, UMC		V_R = 200 V			0.5	µA
1N5617, US		V_R = 400 V			0.5	µA
1N5619, US		V_R = 600 V			0.5	µA
1N5621, US		V_R = 800 V			0.5	µA
1N5623, US		V_R = 1,000 V			0.5	µA
Breakdown voltage	4021	I_R = 50 µA	V_{BR}			
1N5615, US, UMC				220		V
1N5617, US				440		V
1N5619, US				660		V
1N5621, US				880		V
1N5623, US				1100		V
Subgroup 3						
High temperature operation:		T_A = +125°C				
Reverse current leakage	4016	DC method	I_{R2}			
1N5615, US, UMC		V_R = 200 V			55	µA
1N5617, US		V_R = 400 V			55	µA
1N5619, US		V_R = 600 V			55	µA
1N5621, US		V_R = 800 V			55	µA
1N5623, US		V_R = 1,000 V			55	µA

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Reverse recovery time	4031	Condition B1	t_{rr}			
1N5615, US, UMC					150	ns
1N5617, US					150	ns
1N5619, US					250	ns
1N5621, US					300	ns
1N5623, US					500	ns
Capacitance	4001	V_R = 12 V dc, 0.1 < f < 1.0 MHz	C			
1N5615, US, UMC					45	pF
1N5617, US					35	pF
1N5619, US					25	pF
1N5621, US					20	pF
1N5623, US					15	pF
Scope display evaluation	4023	See 4.5.4; n = 116, c = 0				
Subgroup 5						
Not applicable						
Subgroup 6						
Forward surge	4066	Condition A, I_{FSM} = 25 A (pk); ten surges of 8.3 ms each at 1 minute intervals, I_O = 750 mA dc; V_R = rated V_{RWM} (see col. 2 of 1.3); T_A = +100°C; (T_{EC} = +100°C for surface mount devices (US version)).				
Electrical measurement		See table I, subgroup 2 except thermal impedance.				
Subgroup 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Thermal shock	1056	20 cycles, condition D except low temperature shall be achieved using liquid nitrogen (-195°C). Perform a visual for cracked glass. Not required for UMC package.	
Temperature cycling	1051	500 cycles, -65°C to +175°C.	
Hermetic seal	1071		
Electrical measurements		See table I, subgroup 2, and table III, steps 1, 2.	
Subgroup 2			22 devices c = 0
Steady state dc blocking life	1048	1,000 hours; V_R = 80 – 85 percent of rated V_{RWM} (see col. 2 of 1.3).	
Electrical measurements		See table I, subgroup 2 herein except for thermal impedance and table III, step 1. For irradiated devices, include t_{rr} as an end-point measurement.	
Subgroup 4 1/			
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			22 devices c = 0
Barometric pressure, reduced (altitude operation)	1001	Pressure 1N5617, 1N5619 = 8 mm Hg (100,000 ft); 1N5621, 1N5623 = 33 mm Hg (70,000 ft); $V_R = V_{RWM}$ (see col. 2 of 1.3); t = 1 minute (minimum).	

See footnote at end of table.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only - Continued.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 8 1/			45 devices c = 0
Peak reverse power	4065	Peak reverse power (P_{RM})= shall be characterized by the supplier and this data shall be available to the Government. Test shall be performed on each sublot.	
Electrical measurements		During the P_{RM} test, the voltage (V_{BR}) shall be monitored to verify it has not collapsed. Any collapse in V_{BR} during, or after, the P_{RM} test, or rise in leakage current (I_R) after the test that exceeds I_{R1} in table I herein, shall be considered a failure to that level of applied P_{RM}. Progressively higher levels of P_{RM} shall be applied until failure occurs on all devices within the chosen sample size to characterize each sublot.	
Subgroup 9 1/			45 devices c = 0
Resistance to glass cracking	1057	Step stress to destruction by increasing cycles or up to a maximum of 25 cycles, not required for UMC package.	
Subgroup 10			22 devices c = 0
Forward surge	4066	Condition A, I_{FSM} = rated (see col. 6 of 1.3); ten surges of 8.3 ms each at 1 minute intervals superimposed on I_O = 750 mA; $V_R = V_{RWM}$ = (see col. 2 of 1.3); T_A = +100°C.	
Electrical measurement		See table I, subgroup 2.	

1/ The sample size for this step stress requirement shall be determined by the supplier. A statistically significant sample size is required.

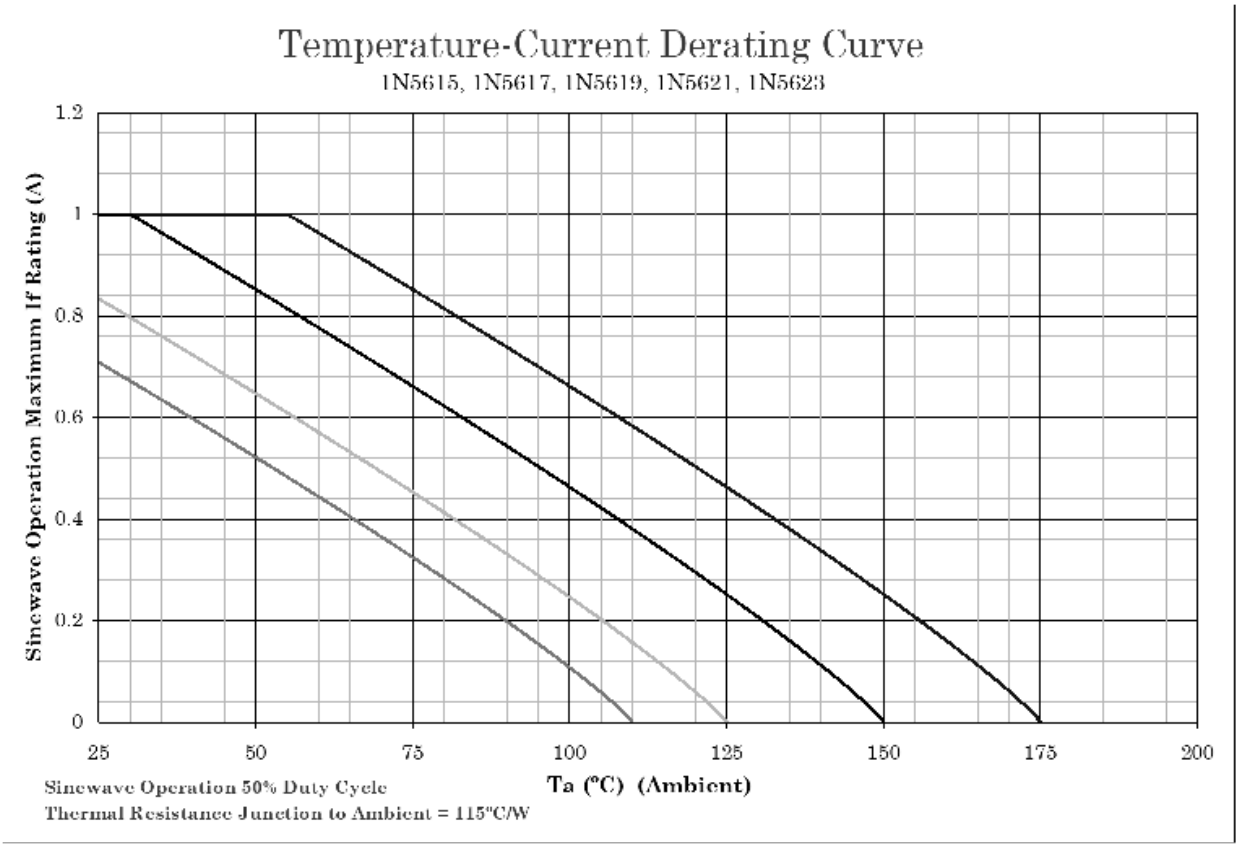
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TABLE III. Groups B, C, and E delta measurements. 1/ 2/ 3/ 4/ 5/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse current leaking change 1N5615, US, UMC 1N5617, US 1N5619, US 1N5621, US 1N5623, US	4016	DC method V_R = 200 V V_R = 400 V V_R = 600 V V_R = 800 V V_R = 1,000 V	ΔI_{R1}		For JAN, JANTX, and JANTXV, ≤ 250 nA dc or 100 percent, whichever is greater; for JANS, ≤ 100 nA dc or 100 percent, whichever is greater.	
2.	Forward voltage change	4011	Condition B, I_F = 3 A dc; pulsed (see 4.5.1) t_p = 300 μs; 2 percent maximum duty cycle.	ΔV_{F1}		±50 mV dc maximum change from previous measured value.	

1/ Devices which exceed the table I, subgroup 2 (group A) limits for this test shall not be accepted.
2/ The delta measurements for group B inspections in table E-VIA (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table III herein, steps 1 and 2.
   b. Subgroup 4, see table III herein, step 2.
   c. Subgroup 5, see table III herein, steps 1 and 2.
3/ The delta measurements for group B inspections in table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table III herein, steps 1 and 2.
   b. Subgroup 6, see table III herein, step 1.
4/ The delta measurements for group C inspections in table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, step 1 (JANS).
   b. Subgroup 6, see table III herein, steps 1 and 2 (JANS), step 1 (JAN, JANTX, and JANTXV).
5/ The delta measurements for group E inspections in table E-IX of MIL-PRF-19500 are: Subgroups 1 and 2, see table III herein, steps 1 and 2.

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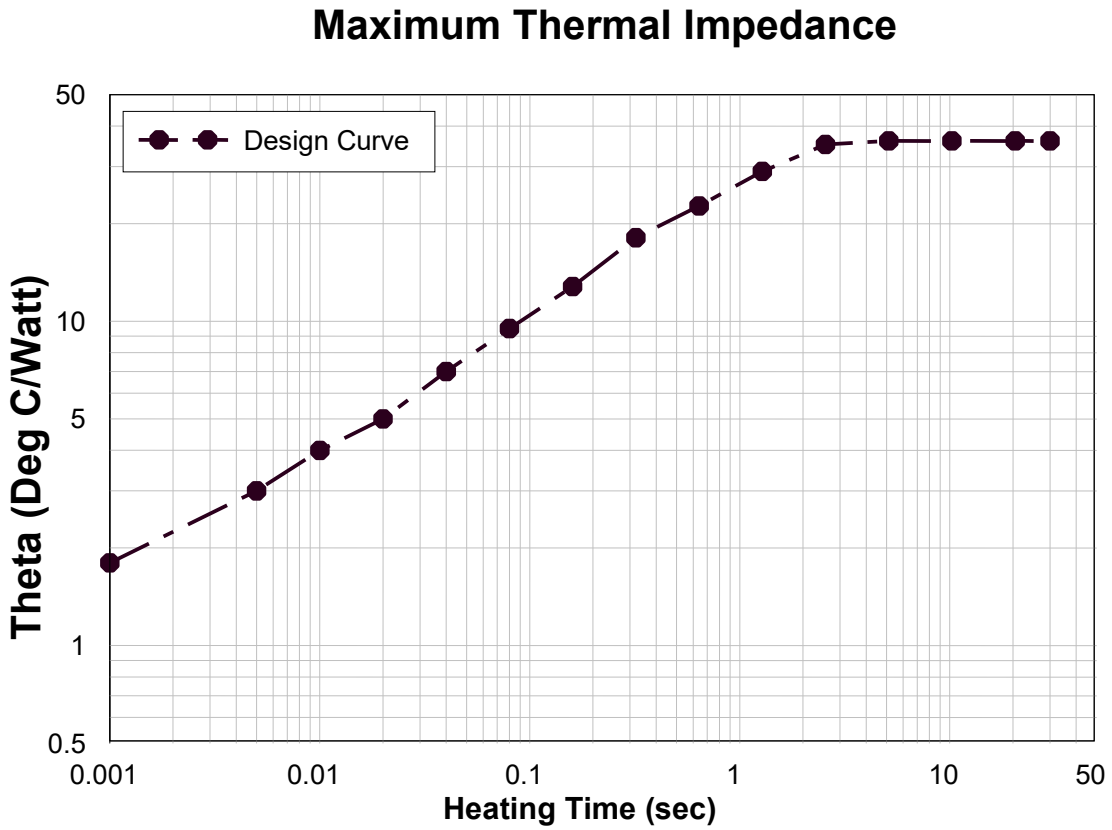


NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed. Also see applications data in section 6.5 for mounting.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ +175°C) and current rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ +125°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤, +150°C, +125°C, and +110°C to show current rating where most users want to limit T_J in their application.

FIGURE 7. Temperature-current derating curve.

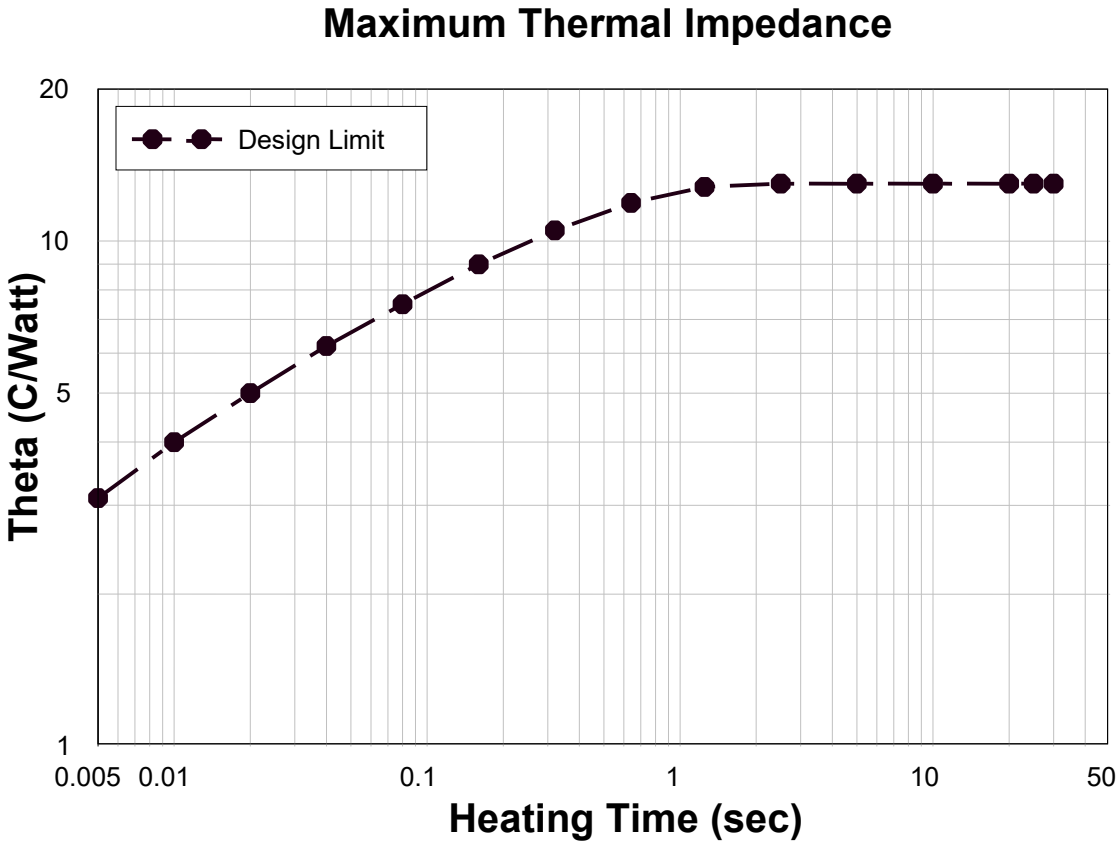
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$Z_{\theta JX}$ = 4°C/W at 10 ms.

FIGURE 8. Thermal impedance curve, $R_{\theta JL}$ = 38°C/W for 1N5615, 1N517, 1N5619, 1N5621, and 1N5623.

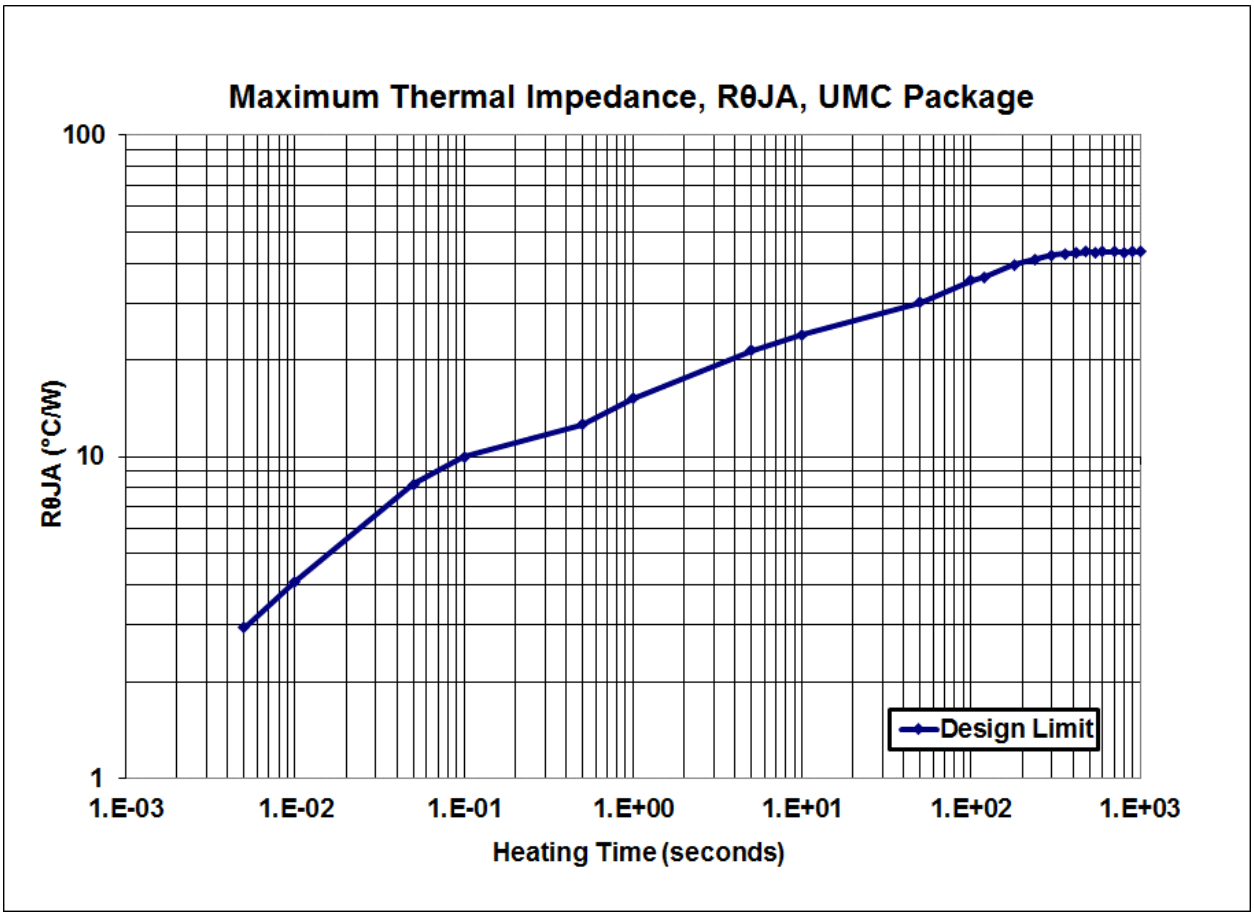
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$Z_{\theta JX}$ = 4°C/W at 10 ms.

FIGURE 9. Thermal impedance curve $R_{\theta JEC}$ = 13°C/W for 1N5615US, 1N517US, 1N5619US, 1N5621US, and 1N5623US.

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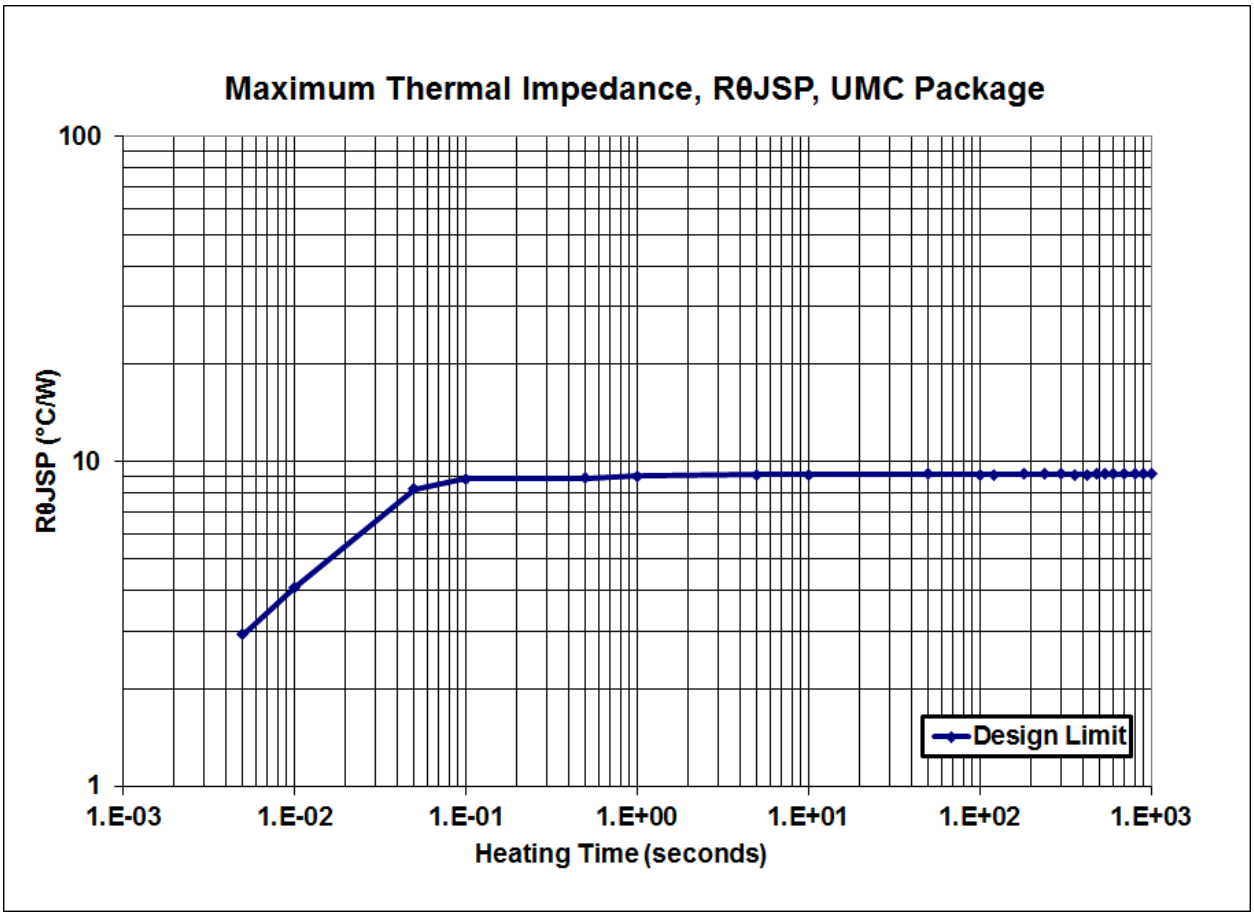


$Z_{\theta JX}$ = 4.08°C/W at 10 ms.

PCB Size: 1.5" x 1.5"
Type: FR4
Copper size: 1.0 oz

FIGURE 10. Thermal impedance curve $R_{\theta JA}$ = 44°C/W.

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$Z_{\theta JX}$ = 4.08°C/W at 10 ms.

PCB Size: 1.5" x 1.5"
Type: FR4
Copper size: 1.0 oz

FIGURE 11. Thermal impedance curve $R_{\theta JSP}$ = 10°C/W.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

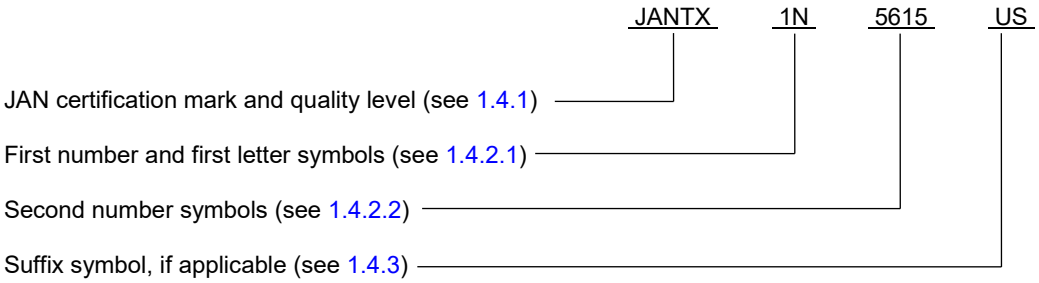
d. The complete PIN, see 1.4 and 6.5.

e. For die acquisition, the JANHC or JANKC letter version should be specified (see figures 4, 5. and 6).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://qpldocs.dla.mil.

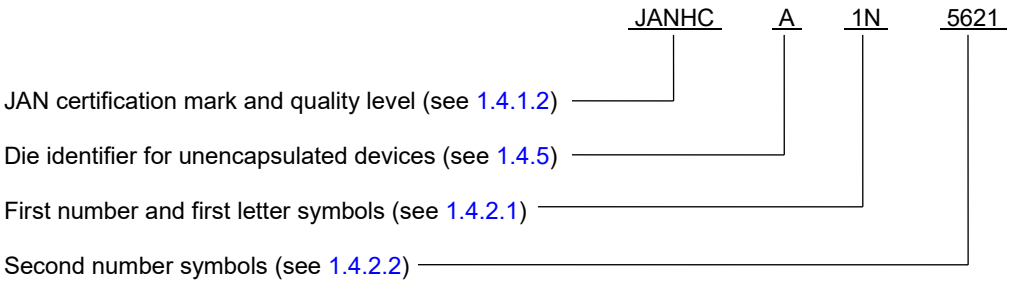
6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs.

6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices in a axial package			
JAN1N5615	JANTX1N5615	JANTXV1N5615	JANS1N5615
JAN1N5617	JANTX1N5617	JANTXV1N5617	JANS1N5617
JAN1N5619	JANTX1N5619	JANTXV1N5619	JANS1N5619
JAN1N5621	JANTX1N5621	JANTXV1N5621	JANS1N5621
JAN1N5623	JANTX1N5623	JANTXV1N5623	JANS1N5623

PINs for devices in a US package			
JAN1N5615US	JANTX1N5615US	JANTXV1N5615US	JANS1N5615US
JAN1N5617US	JANTX1N5617US	JANTXV1N5617US	JANS1N5617US
JAN1N5619US	JANTX1N5619US	JANTXV1N5619US	JANS1N5619US
JAN1N5621US	JANTX1N56201S	JANTXV1N5621US	JANS1N5621US
JAN1N5623US	JANTX1N5623US	JANTXV1N5623US	JANS1N5623US

PINs for devices in a UMC package			
JAN1N5615UMC	JANTX1N5615UMC	JANTXV1N5615UMC	JANS1N5615UMC

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6.5.2 List of PINs for unencapsulated devices. The following is a list of possible PINs available on this specification sheet. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5615) will be identified on the Qualified Manufacturers List (QML).

JANHC and JANKC ordering information			
PIN	Manufacturer		
	14552	33178	34156
1N5615	JANHCA1N5615 JANKCA1N5615	JANHCB1N5615 JANKCB1N5615	JANHCC1N5615 JANKCC1N5615
1N5617	JANHCA1N5617 JANKCA1N5617	JANHCB1N5617 JANKCB1N5617	
1N5619	JANHCA1N5619 JANKCA1N5619	JANHCB1N5619 JANKCB1N5619	
1N5621	JANHCA1N5621 JANKCA1N5621	JANHCB1N5621 JANKCB1N5621	
1N5623	JANHCA1N5623 JANKCA1N5623	JANHCB1N5623 JANKCB1N5623	

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6.6 Applications data.

6.6.1 Half-sine-wave application with 1N5615-1N5623. For a PCB mounting example with FR4 material where the full 1 amp I_O rating (half-sine-wave) is used at a T_J of +175ºC and ambient temperature of +55ºC, the following steps guide the user in what the PCB pad size will need to be with 1 ounce, 2 ounce, and 3 ounce Copper for a 1N5615 to 1N5623. For axial-leaded, the lead length for mounting will be .187 inch (4.76 mm) or less from body to entry point on PCB surface.

a. Use the I_O versus Po curve in figure 12 to look up 1 Amp (X-axis) and follow up to the T_J = 175ºC curve (lower) for 1.04 watts.

b. Calculate maximum thermal resistance needed (175ºC – 55ºC) /1.04 W = 115 ºC/W.

c. Look up thermal resistance of 115ºC/W on Y-axis using a thermal resistance versus pad area plot on one of the three curves in figure 13 for different weights of copper cladding and then intersect curve horizontally to get answer. These curves assume still air, horizontal position.

d. In this example, the answer is: 1 ounce PCB = .042 in X 0.42 in (1.067 mm X 1.067 mm), 2 ounce PCB = .025 in X .025 in (0.635 mm X 0.635 mm), 3 ounce PCB = .014 in X .014 in (0.356 mm X 0.356 mm) for each pad.

e. Add a conservative guard-band to the pad size (larger) to keep T_J below 175ºC.

6.6.2 Square-wave application with 1N5615-1N5623. For a PCB mounting example with FR4 material to support a 0.5 amp I_O square wave switching at a 0.50 duty factor (50 percent duty cycle) at T_J = +125ºC and ambient temperature of +55ºC, the following steps guide the user in what the PCB pad size will need to be with 1 ounce, 2 ounce, and 3 ounce copper.

a. Find size of copper pads on standard FR4 PCB to support operation at 0.5 amp I_O square wave switching at a 0.50 duty factor (50 percent duty cycle) at T_J = +125ºC with T_A = +55ºC.

b. Calculate peak I_F = 0.5A/0.50 duty factor = 1 amp.

c. Use the V_F versus I_F curve in figure 14 to look up I_F = 1 A (Y-axis) and follow across to the T_J = 125ºC curve (middle) for V_F = 0.96 V.

d. Calculate power = I_F * V_F * duty factor = 1 x 0.96 x 0.50 = 0.48 W.

e. Calculate maximum thermal resistance needed (125ºC – 55ºC) / 0.48 W = 146ºC/W.

f. Look up thermal resistance of 146ºC/W on the Y-axis using a thermal resistance versus pad area plot on one of the three curves in figure 14 for different weights of copper cladding and then intersect curve horizontally to get answer. Curves assume still air, horizontal position.

g. Answer: 1 ounce PCB = .02 in X .02 in (0.508 mm X 0.508 mm), 2 ounce PCB = .012 in X .012 in (0.305 mm X 0.305 mm), 3 ounce PCB = .008 in X .008 in (0.202 mm X 0.202 mm) for each pad.

h. A conservative pad guard-band is optional since T_J is only 125ºC. NOTE: Multilayer PCB's, forced air cooling, etc. will improve performance. Closed confinement of the PCB will do the opposite. Please use sound thermal management.

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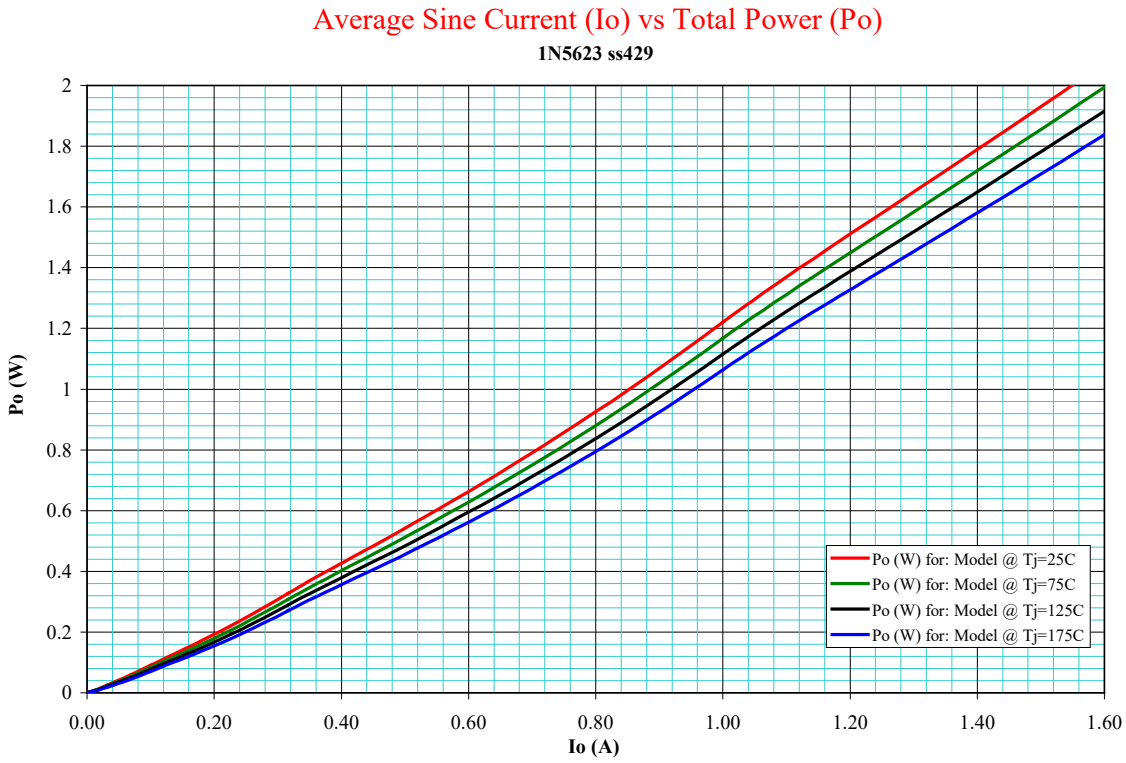


FIGURE 12. Rectifier power versus I_O (average forward current) for 1N5615-1N5623.

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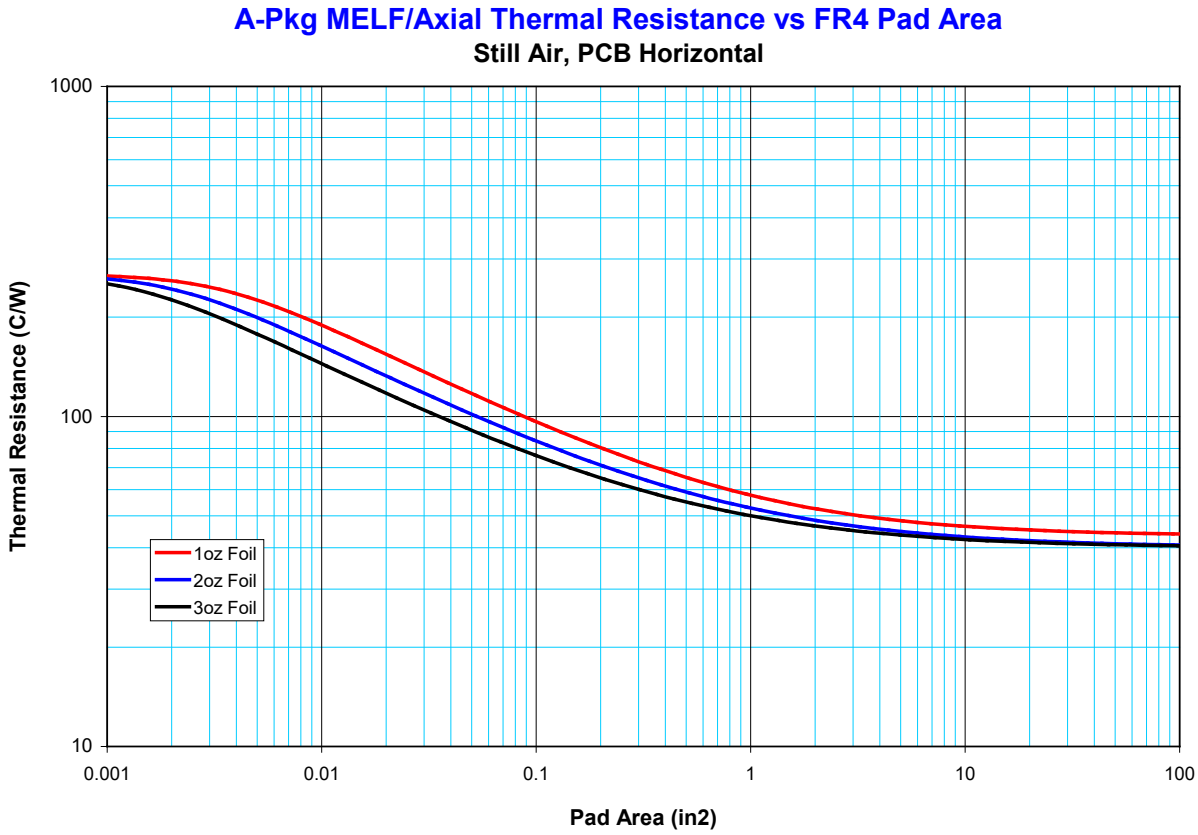


FIGURE 13. Thermal resistance versus pad area (for each pad) with 1, 2, and 3 ounce copper for 1N5615-1N5623.

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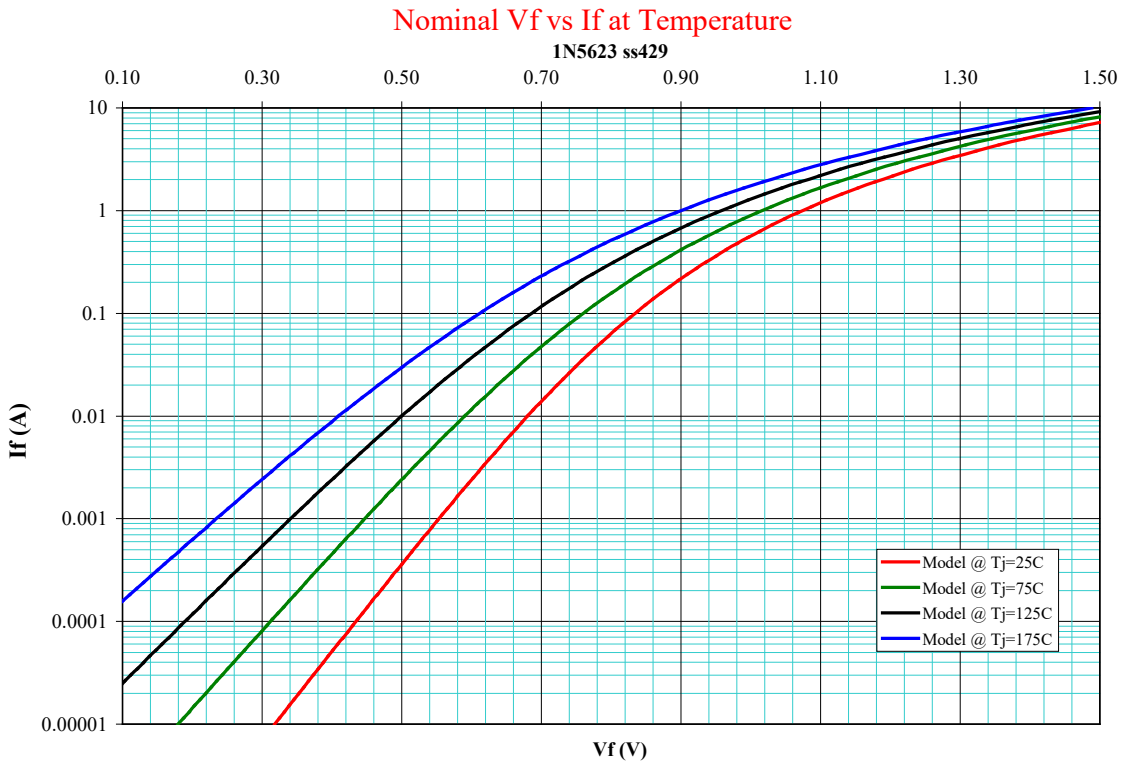


FIGURE 14. Forward voltage versus forward current for 1N5615-1N5623.

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6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

* 6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - EC
Air Force - 85
NASA - NA
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